🔍 Model Comparison Summary

Three classification models — **Decision Tree (Entropy)**, **Logistic Regression**, and **Random Forest** — were trained on the Titanic dataset to predict passenger survival.
All models used a consistent preprocessing pipeline with imputation and one-hot encoding to handle missing and categorical data.

1. **Decision Tree (Entropy)**
   - Uses *information gain* to determine splits.
   - Achieved an accuracy of approximately **79–80%**, with a balanced trade-off between precision and recall.
   - Offers clear interpretability through the decision tree visualization, showing that features such as **sex**, **class**, and **fare** strongly influence survival.
2. **Logistic Regression**
   - A linear baseline model using scaled numeric features.
   - Produced accuracy around **78–79%** with stable performance and fewer signs of overfitting.
   - It effectively captures global patterns but may miss complex, non-linear relationships present in the data.
3. **Random Forest**
   - An ensemble of multiple decision trees combining their outputs.
   - Delivered the highest accuracy (~**83–85%**) and ROC-AUC scores, demonstrating strong generalization and robustness to noise.
   - Although less interpretable than a single Decision Tree, it significantly reduced variance and improved overall performance.

In summary:

- The **Random Forest** model outperformed the other two in predictive accuracy and reliability.
- The **Decision Tree** provided the best interpretability and explanation of key survival factors.
- The **Logistic Regression** model served as a strong linear benchmark with competitive performance.